

NEWINGTON COLLEGE



2015

Assessment 2 (HSC mini)

Year 12 Mathematics – Extension 1

General Instructions:

- Date of task - Monday 30th March (Wk 10B)
- Reading time – 5 mins
- Working time – 120 mins
- Weighting - 30%
- BOSTES approved calculators may be used.
- A table of standard integrals is provided at the back of the paper.
- Attempt all questions.
- Show all relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I (10 marks)

- Answer questions 1 to 10 on the multiple choice answer sheet provided at the end of this paper.
- Allow about 15 minutes for this section.

Section II (60 marks)

- Answer questions 11 to 14 on the writing paper provided.
- **Start each question in a new writing booklet.**
- Each page must show the candidate's computer number.

Outcomes to be assessed:

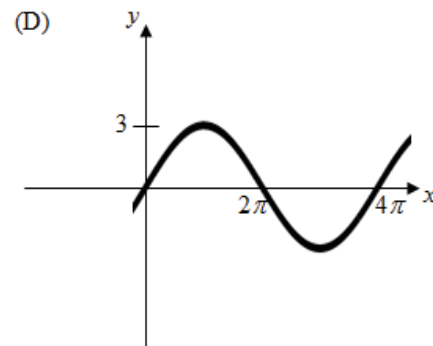
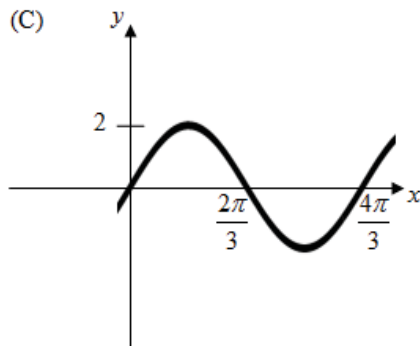
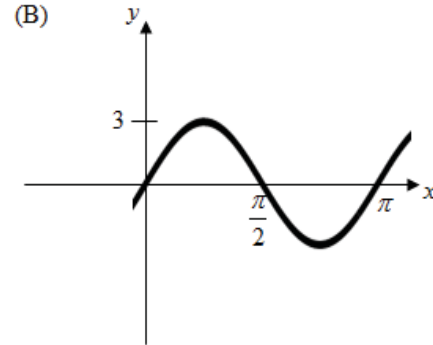
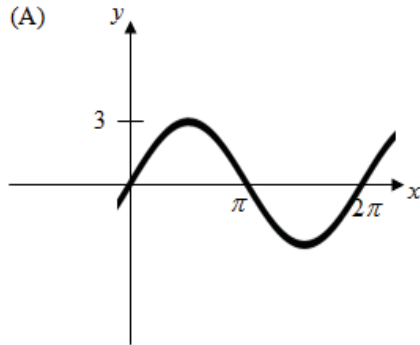
- HE1** Appreciates interrelationships between ideas drawn from different areas of mathematics.
- HE4** Uses the relationship between functions, inverse functions and their derivatives.
- HE6** Determines integrals by reduction to a standard form through a given substitution.

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Section I**10 Marks**

Attempt Questions 1-10 on the multiple choice answer sheet.
Allow about 15 minutes for this section.

- 1) Which curve represents $y = 3\sin 2x$?



- 2) The acute angle (*to the nearest degree*) between two lines which have gradients of $\frac{1}{3}$ and $\frac{-2}{3}$ is:

- (A) 1° (B) 23°
(C) 39° (D) 52°

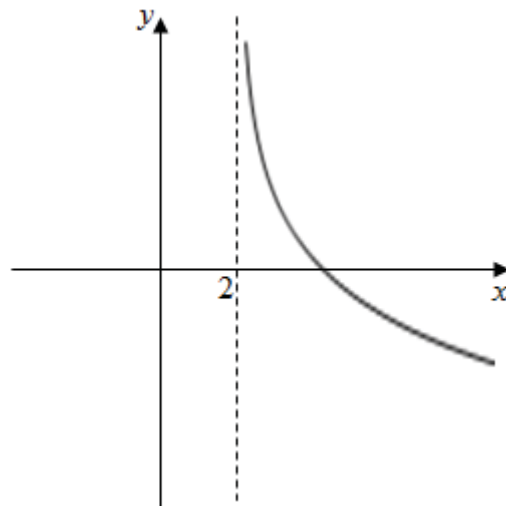
- 3) What is a possible equation of the following curve?

(A) $y = \ln(x+2)$

(B) $y = \ln(x-2)$

(C) $y = \ln\left(\frac{1}{x+2}\right)$

(D) $y = \ln\left(\frac{1}{x-2}\right)$



- 4) Find $\int \cos x e^{\sin x} dx$
- (A) $\sin x e^{\sin x} + C$
- (B) $\cos^2 x e^{\sin x} + C$
- (C) $e^{\sin x} + C$
- (D) $\cos x e^{\sin x} + C$
- 5) Which function below only has an inverse function if the domain of $f(x)$ is restricted?
- (A) $f(x) = \sin^{-1} x$
- (B) $f(x) = \tan^{-1} x$
- (C) $f(x) = \cos x$
- (D) $f(x) = \log_e x$
- 6) $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x}$ is equal to:
- (A) 0
- (B) 1
- (C) $\frac{3}{2}$
- (D) $\frac{2}{3}$
- 7) The expansion of $\cos(A + B)$ is equal to:
- (A) $\sin A \cos B - \cos A \sin B$
- (B) $\cos A \cos B - \sin A \sin B$
- (C) $\sin A \cos B + \cos A \sin B$
- (D) $\cos A \cos B + \sin A \sin B$

8) The exact value of $\sin^{-1}\left(\cos\frac{2\pi}{3}\right)$ is:

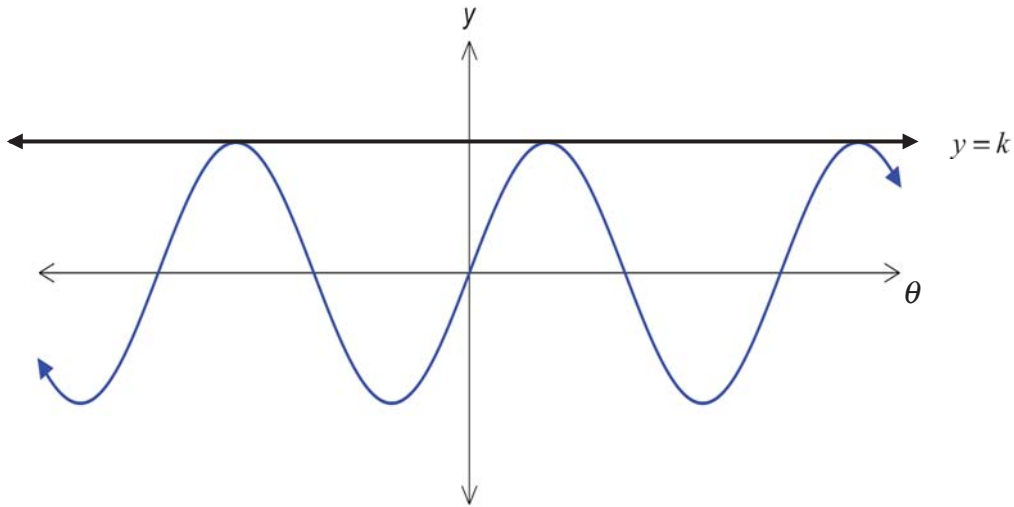
(A) $\frac{\pi}{6}$

(B) $\frac{2\pi}{3}$

(C) $-\frac{\pi}{6}$

(D) $-\frac{2\pi}{3}$

9) The curve $y = 2\sin 2\theta$ and the horizontal line $y = k$ are drawn below.



The solutions of the equation $2\sin 2\theta - k = 0$ as shown on this diagram are:

(A) $\theta = -\frac{3\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}$

(B) $\theta = -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

(C) $\theta = -\frac{7\pi}{12}, \frac{\pi}{12}, \frac{5\pi}{12}$

(D) $\theta = -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$

10) $\int \frac{dx}{4+9x^2} =$

(A) $\frac{1}{3}\tan^{-1}\frac{2x}{3}$

(B) $\frac{1}{6}\tan^{-1}\frac{3x}{2}$

(C) $\frac{1}{6}\tan^{-1}\frac{2x}{3}$

(D) $\frac{1}{3}\tan^{-1}\frac{3x}{2}$

Section II**Attempt questions 11-14****Allow about 1 hour and 45 minutes for this section.****Question 11 (16 Marks)- Use a SEPARATE writing booklet.**

a) Differentiate:

$$(i) \quad y = \cos(x^2 - 3) \quad \mathbf{1}$$

$$(ii) \quad f(x) = \tan^{-1}\left(\frac{5x}{4}\right) \quad \mathbf{2}$$

$$b) \quad \text{Prove } \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} = \frac{2}{\sin 2x} \quad \mathbf{3}$$

$$c) \quad \text{Find } \int \frac{\cos x}{\sin x} dx \quad \mathbf{1}$$

$$e) \quad \text{Find } \int \frac{1}{2} \sin 2x \sec x dx \quad \mathbf{2}$$

$$f) \quad \text{Find the exact value of } \sin 105^\circ \quad \mathbf{3}$$

$$g) \quad \text{If } \sin \theta = \frac{3}{5} \text{ and } \theta \text{ is acute, find the exact value(s) of } \tan \frac{\theta}{2} \quad \mathbf{4}$$

Question 12 (15 Marks)- Use a SEPARATE writing booklet.

- a) Evaluate $\int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$, leaving your answer in exact form. **3**
- b) Evaluate $\int_0^{\frac{\pi}{2}} 3\sin 3\theta \, d\theta$ **2**
- c) Find $\frac{d}{dx} \cos^{-1} x^2$ **2**
- d) Find the general solution to $2\sin x = \sqrt{3}$,
expressing your answer in terms of π . **2**
- e) Solve $1 + 2\cos^2 x = 5\sin x$, $0 \leq x \leq 2\pi$ **2**
- f) (i) Express $\sqrt{3}\cos\theta - \sin\theta$ in the form $r\cos(\theta + \alpha)$, where
 $r > 0$ and $0 < \alpha < \frac{\pi}{2}$, giving r and α as exact values. **3**
- (ii) Evaluate the minimum value of the expression $\sqrt{3}\cos\theta - \sin\theta$. **1**

Question 13 (15 Marks)- Use a SEPARATE writing booklet.

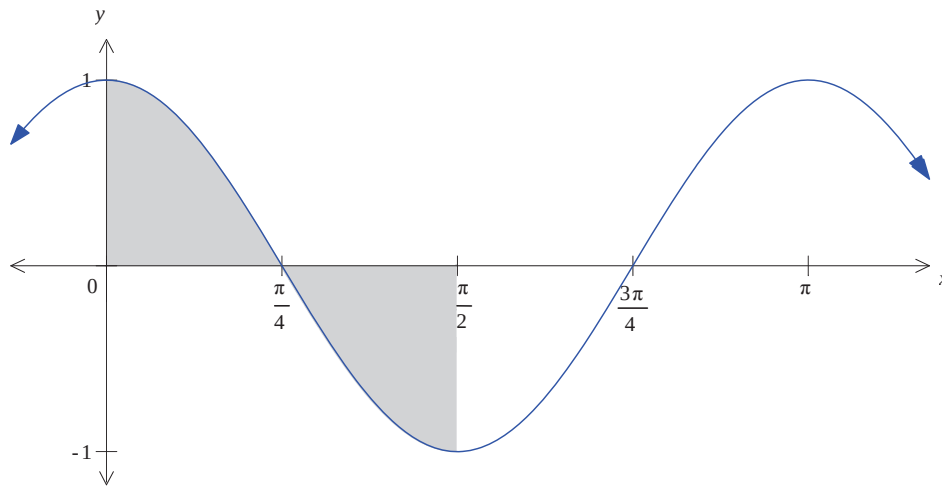
- a) Given the function $f(x) = e^x + 2$
- (i) Write an expression for the inverse function $f^{-1}(x)$ **2**
 - (ii) Write down the domain and range of $f^{-1}(x)$ **2**
- b) Given the function $y = \frac{\ln x}{x}$
- (i) Find any x and y intercepts **1**
 - (ii) Find any stationary points and determine their nature **3**
 - (iii) Clearly sketch the curve, indicating all the key features detailed above, as well as any asymptotes. **2**
(You do not have to show any points of inflexion)
- c) Consider the function $f(x) = 3\cos^{-1}(x+1)$
- (i) State the domain and range of this function. **2**
 - (ii) Draw a neat sketch of this function, showing this information. **1**
 - (iii) Find the equation of the tangent to this curve at the point $\left(-1, \frac{3\pi}{2}\right)$ **2**

Question 14 (14 Marks)- Use a SEPARATE writing booklet.

a) Show that $\frac{d}{dx}(x \cdot \sin^{-1} 3x) = \sin^{-1}(3x) + \frac{3x}{\sqrt{1-9x^2}}$ 2

Hence, or otherwise, find $\int \sin^{-1} 3x \, dx$ 4
 (Hint: use the substitution $u = 1 - 9x^2$)

- b) The area bounded by the curve $y = \cos 2x$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x -axis. 3



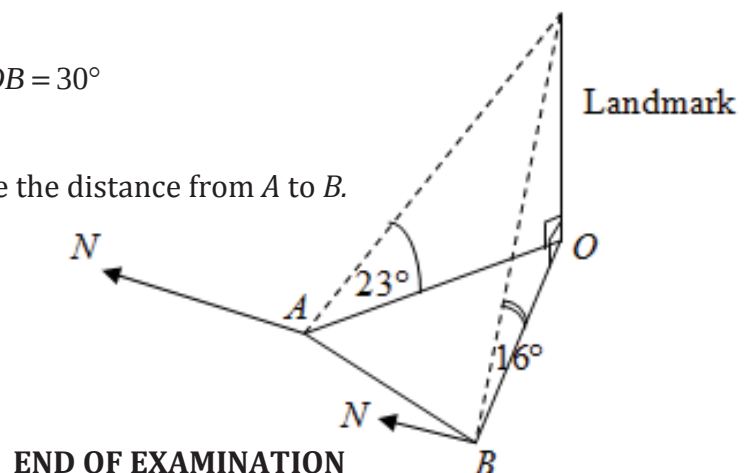
Find, in exact form, the volume of the solid of revolution formed.

- c) The bearing of a vertical landmark from point A is 139° and the angle of elevation of the landmark from A is 23° . From point B the same landmark has a bearing of 109° and an angle of elevation of 16° . The landmark is 150m tall.

(i) Show that $AO = 150 \cot 23^\circ$ 1

(ii) Show that $\angle AOB = 30^\circ$ 2

(iii) Hence, calculate the distance from A to B . 2



END OF EXAMINATION

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Student Number :

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SECTION I – Multiple Choice Answer Sheet

*Instructions – 1. Tear off this page and write your student number in box above.
2. Colour in the circle corresponding to your correct answer.*

- | | | | | |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Question 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| Question 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Section 1

1) Amp = 3 Period = $\frac{2\pi}{2} = 2$ (B)

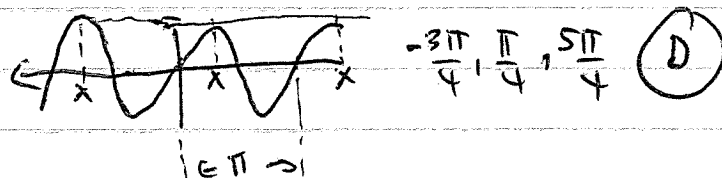
8) $\sin^{-1}\left(\cos \frac{2\pi}{3}\right) = \sin^{-1}\left(-\frac{1}{2}\right)$
 $= -\sin^{-1}\left(\frac{1}{2}\right)$

2) $\tan \theta = \frac{\frac{1}{3} - \frac{2}{3}}{1 + \frac{1}{3} \times \frac{2}{3}}$
 $\theta = \tan^{-1}\left[\frac{1}{7}\right] = \tan^{-1}\left[\frac{9}{7}\right]$
 $\approx 52^\circ$ (D)

(C) $= -\frac{\pi}{6}$

9) Period = $\frac{2\pi}{2} = \pi$

3) Upside down shifted 2 to right.



$\therefore y = -\ln(x-2)$

$y = \ln(x-2)^{-1}$

$y = \ln\left(\frac{1}{x-2}\right)$ (D)

10) $\int \frac{dx}{4+9x^2} = \frac{1}{9} \int \frac{dx}{\frac{4}{9} + x^2}$

4) In the form $\int f'(x)e^{f(x)} dx = e^{f(x)} + c$
 (C)

(B)

$= \frac{1}{9} \times \frac{1}{\frac{2}{3}} \tan^{-1} \frac{x}{\frac{2}{3}} + c$

$= \frac{1}{6} \tan^{-1} \frac{3x}{2} + c$

5) more than 1 x
 for 1 y
 $y = \cos x$ (C)

6) $\frac{2}{3} \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = \frac{2}{3}$ (D)

7) (B)

Question 11

$$\begin{aligned} \text{a) i) } y &= \cos(x^2 - 3) \\ \frac{dy}{dx} &= -2x \sin(x^2 - 3) \end{aligned}$$

$$\text{ii) } f(x) = \tan^{-1} \frac{5x}{4}$$

$$\begin{aligned} f'(x) &= \frac{\frac{5}{4}}{1 + \left(\frac{5x}{4}\right)^2} = \frac{5}{4 + 25x^2} \\ &= \frac{20}{16 + 25x^2} \end{aligned}$$

$$\begin{aligned} \text{b) LHS} &= \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \\ &= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} \\ &= \frac{1}{\sin x \cos x} \\ &= \frac{2}{2 \sin x \cos x} \\ &= \frac{2}{\sin 2x} = \text{RHS} \end{aligned}$$

$$\text{c) } \int \frac{\cos x}{\sin x} dx = \ln |\sin x| + C$$

$$\begin{aligned} \text{d) } \int \frac{1}{2} \sin 2x \sec x dx &= \int \frac{1}{2} \frac{2 \sin x \cos x}{\cos x} dx \\ &= \int \sin x dx \\ &= -\cos x + C \end{aligned}$$

$$\begin{aligned} \text{e) } \sin(105) &= \sin(45 + 60) \\ &= \sin 60 \cos 45 + \cos 60 \sin 45 \\ &= \frac{\sqrt{3}}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{f) } \sin \theta &= \frac{3}{5} \\ \therefore \frac{3}{5} &= \frac{2t}{1+t^2} \end{aligned}$$

$$3 + 3t^2 = 10t$$

$$3t^2 - 10t + 3 = 0$$

$$(3t-1)(t-3) = 0$$

$$t = \frac{1}{3} \quad \text{or} \quad t = 3$$

$$\tan \frac{\theta}{2} = \frac{1}{3} \quad \text{or} \quad \tan \frac{\theta}{2} = 3$$

$$\theta \text{ is acute } \therefore \tan \frac{\theta}{2} = \frac{1}{3}$$

Question 12

$$\begin{aligned}
 a) \int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} &= \sin^{-1}\left(\frac{x}{2}\right) \Big|_1^{\sqrt{3}} \\
 &= \sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}\frac{1}{2} \\
 &= \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 b) \int_0^{\frac{\pi}{2}} 3 \sin 3\theta \, d\theta &= \left[-\cos 3\theta \right]_0^{\frac{\pi}{2}} \\
 &= -\cos \frac{3\pi}{2} - (-\cos 0) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 c) \frac{d}{dx} \cos^{-1} x^2 & \quad u = x^2 \\
 & \quad \frac{du}{dx} = 2x \\
 \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\
 &= \frac{-1}{\sqrt{1-u^2}} \times 2x \\
 &= \frac{-2x}{\sqrt{1-x^4}}
 \end{aligned}$$

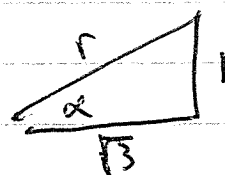
$$\begin{aligned}
 d) \sin x &= \frac{\sqrt{3}}{2} \\
 x &= \frac{\pi}{3}, \pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3} \\
 x &= n\pi + (-1)^n \frac{\pi}{3}
 \end{aligned}$$

$$\begin{aligned}
 e) 1 + 2\cos^2 x &= 5\sin x \\
 2\cos^2 x - 5\sin x + 1 &= 0 \\
 2(1 - \sin^2 x) - 5\sin x + 1 &= 0 \\
 2 - 2\sin^2 x - 5\sin x + 1 &= 0 \\
 2\sin^2 x + 5\sin x - 3 &= 0 \\
 (2\sin x - 1)(\sin x + 3) &= 0 \\
 \sin x &= \frac{1}{2} \quad \sin x = -3 \\
 \Downarrow & \quad \text{NO SOLN}
 \end{aligned}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned}
 f) \sqrt{3} \cos \theta - \sin \theta &= r \cos(\theta + \alpha) \\
 &= r \cos \theta \cos \alpha - r \sin \theta \sin \alpha
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{3} &= r \cos \alpha & 1 &= r \sin \alpha \\
 \cos \alpha &= \frac{\sqrt{3}}{r} & \sin \alpha &= \frac{1}{r}
 \end{aligned}$$



$$\begin{aligned}
 r &= 2 \\
 \alpha &= \frac{\pi}{6}
 \end{aligned}$$

$$\therefore \sqrt{3} \cos \theta - \sin \theta = 2 \cos\left(\theta + \frac{\pi}{6}\right)$$

$$ii) \text{ min value } = -2$$

Question 13

1) f: $y = e^x + 2$

f^{-1} : $x = e^y + 2$

$e^y = x - 2$

$y = \log_e(x - 2)$

i) D: $x > 2$

R: all real y

2) $y = \frac{\ln x}{x}$

(i) x int: $y = 0$

$0 = \frac{\ln x}{x}$

$\ln x = 0 \therefore x = 1$

no y intercept

(ii) $y' = \frac{\frac{1}{x} \times x - 1 \times \ln x}{x^2}$

$= \frac{1 - \ln x}{x^2}$

4 Pts: $y' = 0$

$0 = 1 - \ln x$

$\ln x = 1 \therefore x = e$

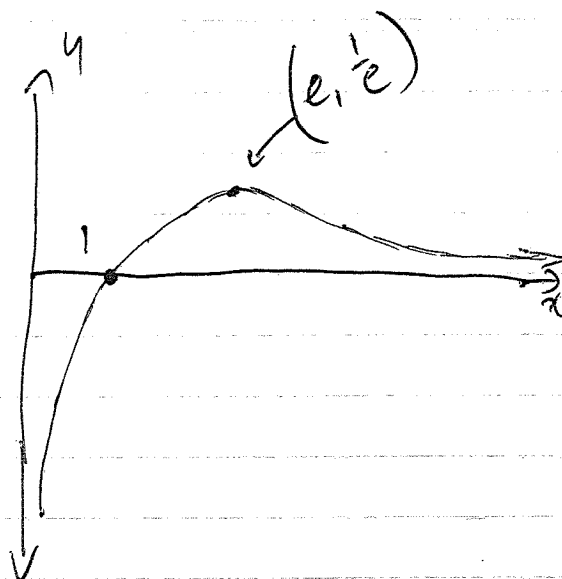
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x	2	e	3
y'	$\frac{1 - \ln 2}{4}$	0	$\frac{1 - \ln 3}{9}$

/ — \

$\therefore$ max at $(e, \frac{1}{e})$

iii)



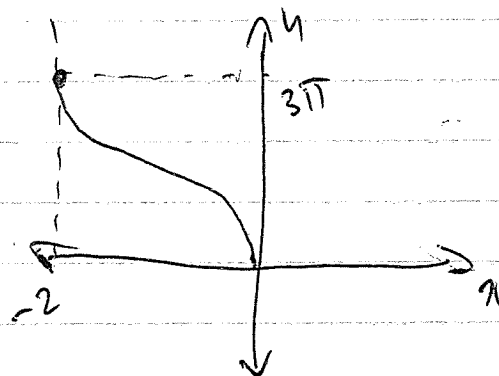
c) $f(x) = 3\cos^{-1}(x+1)$

i) D: $-1 \leq x+1 \leq 1$

$-2 \leq x \leq 0$

R: $0 \leq y \leq 3\pi$

ii)



iii) $f(x) = 3\cos^{-1}(x+1)$

$f'(x) = \frac{-1}{\sqrt{1-(x+1)^2}}$

$f'(-1) = \frac{-3}{1} = -3$

$y - \frac{3\pi}{2} = -3(x+1)$

$3x + y - \frac{3\pi}{2} + 3 = 0$

$3x + y - \frac{3\pi}{2} + 3 = 0$

4/6

Question 14

$$1) \frac{d}{dx} (x \sin^{-1} 3x)$$

$$= \frac{x}{\sqrt{\left(\frac{1}{3}\right)^2 - x^2}} + \sin^{-1} 3x$$

$$= \frac{3x}{\sqrt{1-9x^2}} + \sin^{-1} 3x$$

$$i) \int \frac{3x}{\sqrt{1-9x^2}} + \sin^{-1} 3x \, dx = x \sin^{-1} 3x$$

$$\therefore \int \sin^{-1} 3x \, dx = x \sin^{-1} 3x - \int \frac{3x}{\sqrt{1-9x^2}} \, dx$$

①

$$u = 1-9x^2$$

$$\frac{du}{dx} = -18x, \, dx = \frac{du}{-18x}$$

$$\textcircled{1} \Rightarrow \int \frac{3x}{\sqrt{u}} \cdot \frac{du}{-18x}$$

$$= -\frac{1}{6} \int \frac{du}{\sqrt{u}}$$

$$= -\frac{1}{6} \cdot 2\sqrt{u}$$

$$= x \sin^{-1} 3x + \frac{1}{3} \sqrt{u}$$

$$= x \sin^{-1} 3x + \frac{1}{3} \sqrt{1-9x^2} + C$$

$$b) V = \pi \int_0^{\frac{\pi}{2}} y^2 \, dx \quad y^2 = \cos^2 2x$$

$$= \pi \int_0^{\frac{\pi}{2}} \cos^2 2x \, dx$$

$$\cos 2x = 2\cos^2 x - 1$$

$$\therefore \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 4x \, dx$$

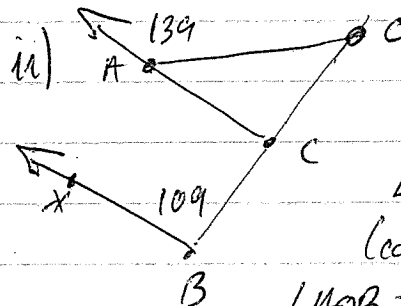
$$= \frac{\pi}{2} \left[x + \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{2} + \frac{1}{4} \sin 4 \cdot \frac{\pi}{2} \right) - \left(0 + \frac{1}{4} \sin 0 \right) \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{2} + 0 \right] = \frac{\pi^2}{4}$$

$$c) i) \tan 23 = \frac{150}{AO}$$

$$\therefore AO = 150 \cot 23$$



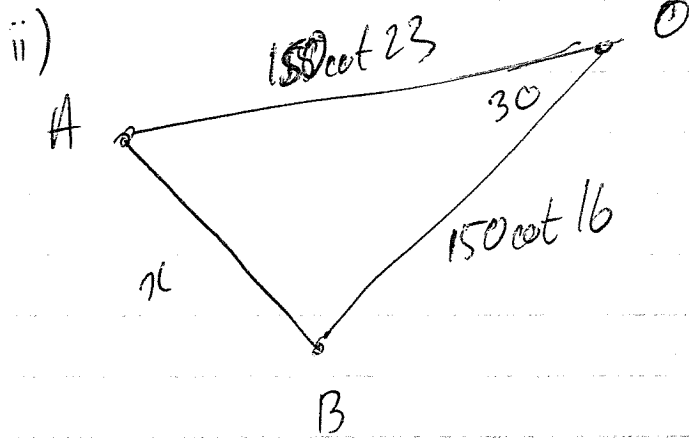
$$\angle AOC = 139^\circ$$

(corresponding to $\angle CBX$)

$$\angle AOB = 139 - 109$$

(exterior angle)

$$\angle AOB = 30^\circ$$



$$x^2 = (150 \cot 23)^2 + (150 \cot 16)^2 - 2 \cdot 150^2 \cot 23 \cdot \cot 16 \cdot \cos 30$$

$$x = 150 \sqrt{\frac{1}{\tan^2 23} + \frac{1}{\tan^2 16} - \frac{2 \cos 30}{\tan 23 \tan 16}}$$

$$x = 279.896 \text{ m}$$